

## Object Storage- Client (S3 browser)

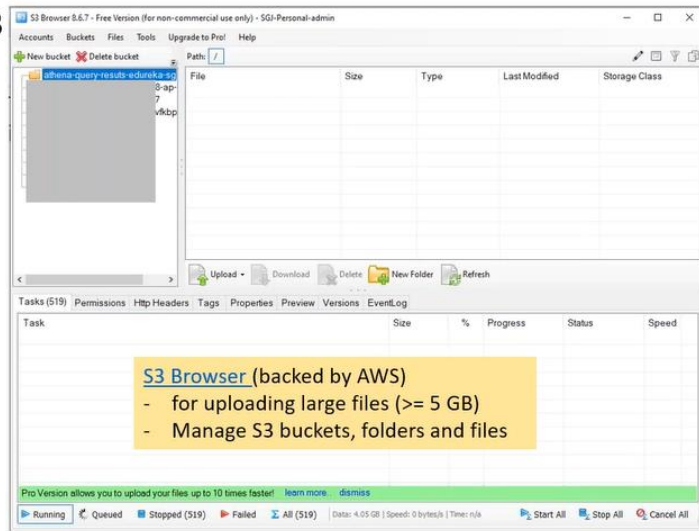
Encrypt the data, transfer the data through public internet.

- Simple Storage Service – S3

- Object Storage
- Bucket
- Folder is an illusion
- Different storage levels (save \$\$)
- Versioning
- Static site hosting
- Events
- Lifecycle Management

[Mount S3 bucket on EC2 Linux Instance](#)

[Mount S3 Bucket as a Windows Drive](#)

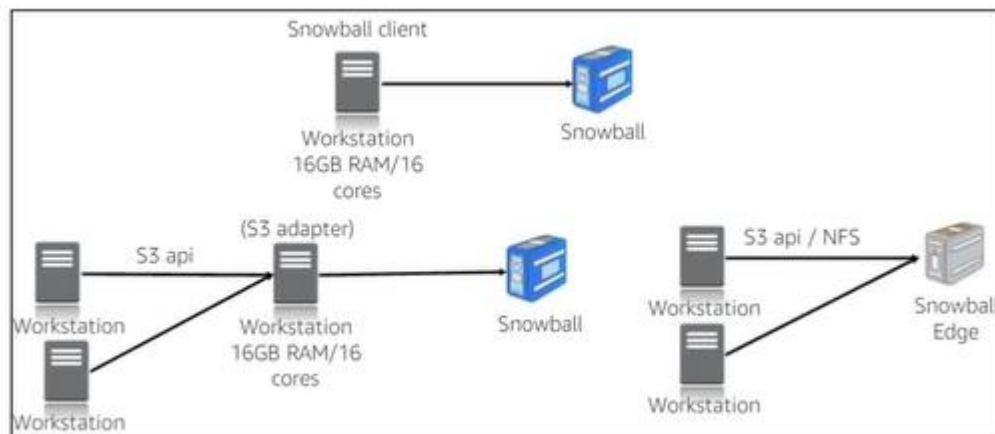


mount S3 bucket on EC2 instance (TNT Drive).

## Object Storage- Client (WinSCP)

Using the script, automate the data transfer.

## Snow Family



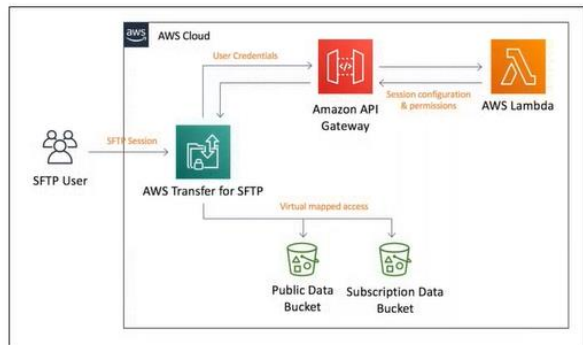
**Helps** eliminate challenges that can be encountered with large-scale data transfers including high network costs, long transfer time and security concerns.

Use for import data to cloud or export data from cloud at computing locations.

- Snowcone
- Snowball Edge

- Snow Mobile

## AWS Transfer Family (SFTP,FTPS,FTP)



### Benefits

Reduce operational burdens

Streamline workflow migrations

Integrate processing and management

API can be enabled on service for programmatic file transfers

If S3 is used to host static web site, this transfer service can help you manage contents without exposing AWS console to the third party or resource involved

- SFTP, FTPS, FTP (exchange data on cloud)
- SFTP: SSL (end point ssl)
- FTPS: TLS
- FTP: non secure

step1: create a server.

step2: select the protocol (SFTP).

step3: Choose an Identity provider (service managed).

step4: Choose an end point

step5: Choose a domain

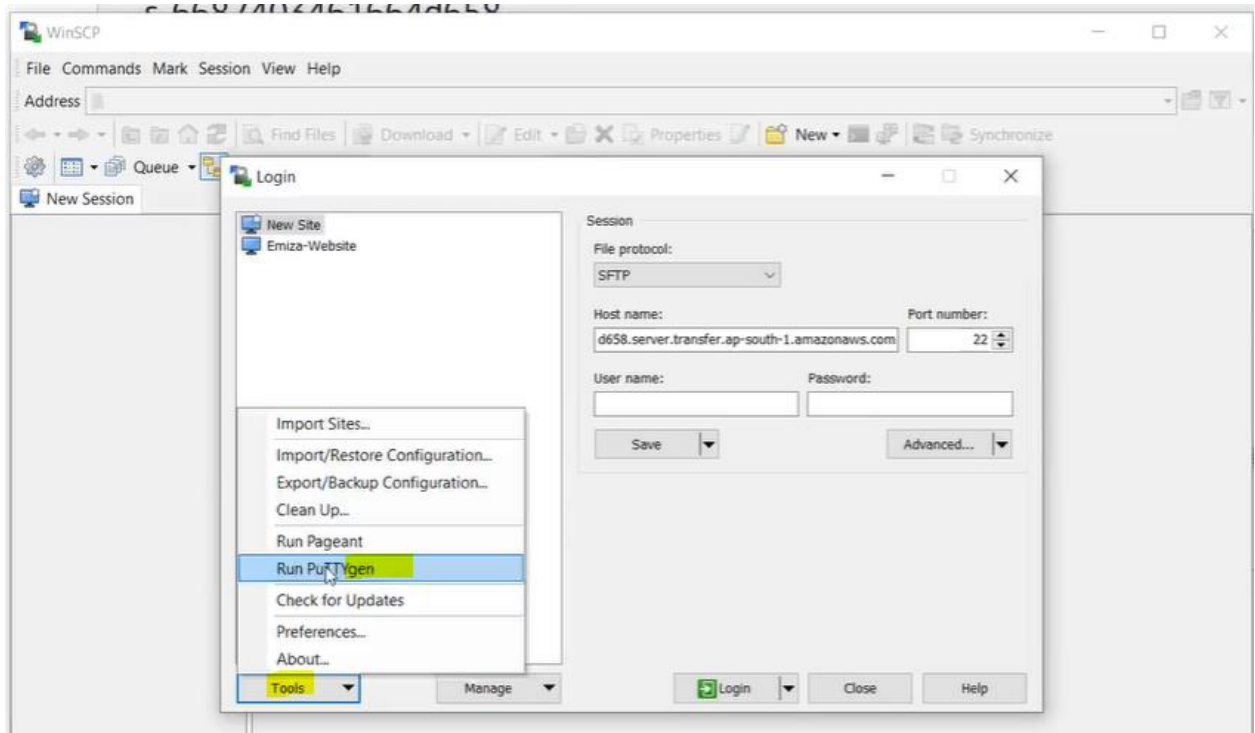
step6: Configure additional details

### How app securely communicate?

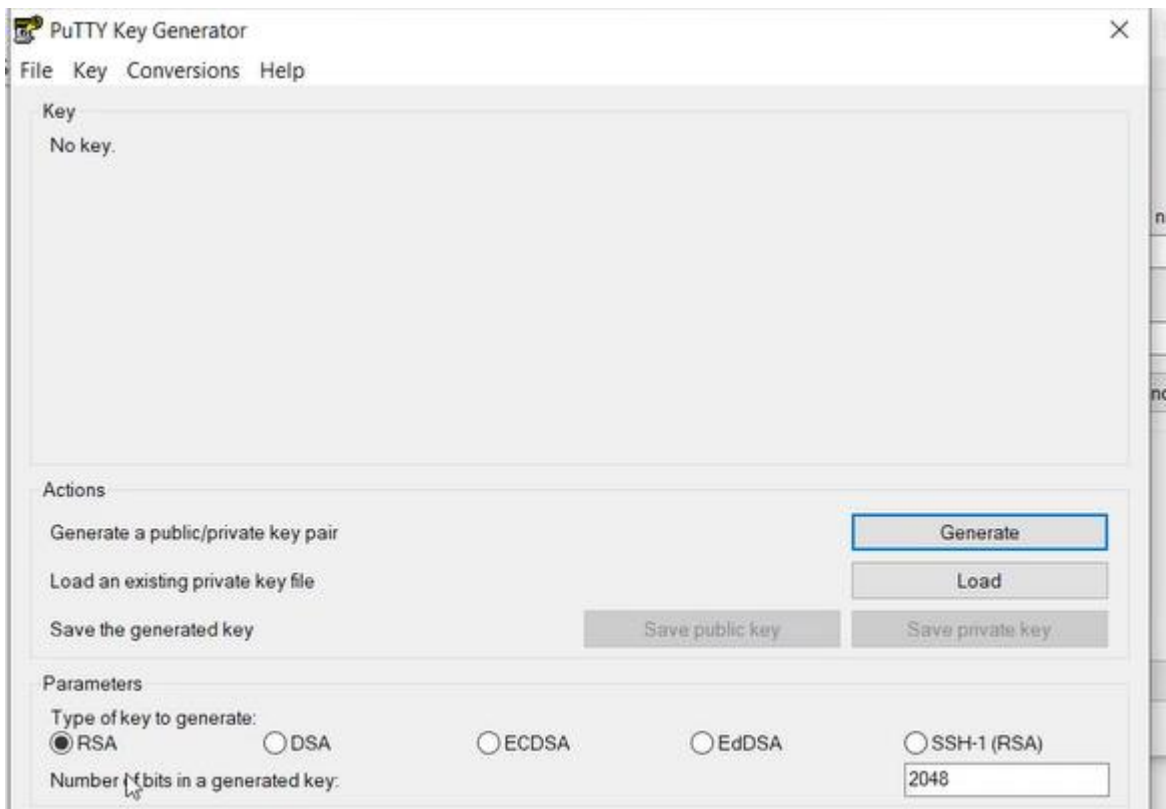
Concept: Encrypt Data using the public key can be only decrypted using corresponding private key.

step7: Connect SFTP server through WinSCP

We have to create a user and every user carry a certificate, certificate is a public/private key generated through the putty key generator



- click on generate button (move key randomly to generate key)



- save public key ( RSA 2048)
- save private key ( RSA 2048)

step8: click on SFTP server

- Add User
  - step8.1: create a role for Transfer (S3Full Access)
  - User Name
  - IAM role – sftp role
  - Home Directory ? S3 bucket (dump the data)
  - SSH public key- copy the public key
  - Click on Add button

WinSCP:

connect to the SFTP server.

- host: SFTP end point
- user name:
- password: (Advanced)
- password-> authenticate-> private key
- click on login

copy the files, available is S3 bucket.

How to retain timestamp?

Tools-> preferences->transfer-> default-> edit (uncheck set permission,preserve time stamp)

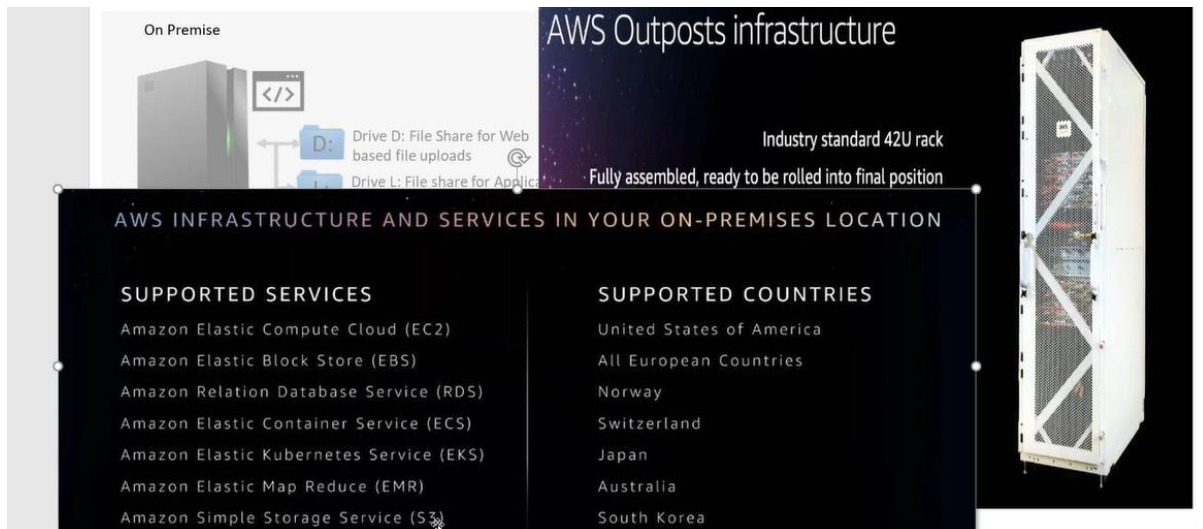
**Diff between DataSync and S3 Transfer Acceleration?**

- DataSync provides two way communication, can switch the endpoints
- DataSync also provides validation feature.
- S3 Transfer Acceleration also don't ensure about data integrity.

## UC:

- Direct Connect is so costly \$.
- Not sure if we can operate in cloud.
  - Need to explore
- Not comfortable transferring data on the public internet.
- What would be the net benefit post migration?

## Outposts:



The image shows a diagram of AWS Outposts infrastructure. On the left, a section titled 'On Premise' shows a server rack with a code icon and labels for 'Drive D: File Share for Web based file uploads' and 'Drive L: File share for Application'. To the right, a large white server rack is shown with the text 'AWS Outposts infrastructure' and 'Industry standard 42U rack'. Below this, it says 'Fully assembled, ready to be rolled into final position'. A central dark box contains the title 'AWS INFRASTRUCTURE AND SERVICES IN YOUR ON-PREMISES LOCATION' and two columns of supported services and countries.

**AWS INFRASTRUCTURE AND SERVICES IN YOUR ON-PREMISES LOCATION**

| SUPPORTED SERVICES                      | SUPPORTED COUNTRIES      |
|-----------------------------------------|--------------------------|
| Amazon Elastic Compute Cloud (EC2)      | United States of America |
| Amazon Elastic Block Store (EBS)        | All European Countries   |
| Amazon Relation Database Service (RDS)  | Norway                   |
| Amazon Elastic Container Service (ECS)  | Switzerland              |
| Amazon Elastic Kubernetes Service (EKS) | Japan                    |
| Amazon Elastic Map Reduce (EMR)         | Australia                |
| Amazon Simple Storage Service (S3)      | South Korea              |

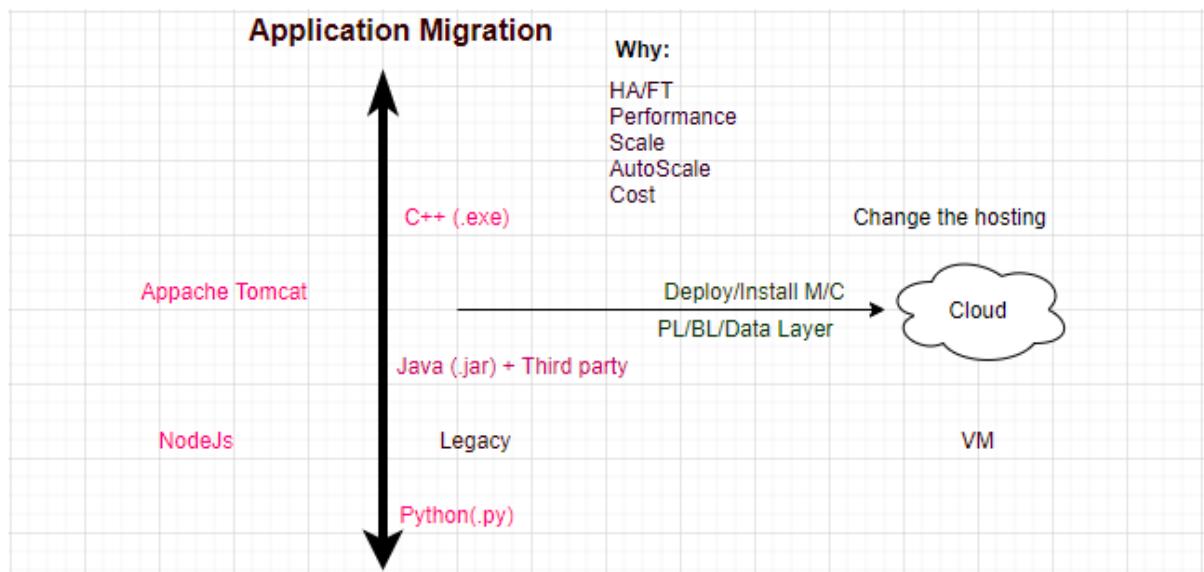
## AWS remote work and learning:

- Tools for employees to work securely from any location
- AWS remote work and learning solutions allow employees, contact center agents, and students to remain productive and connected while working from home

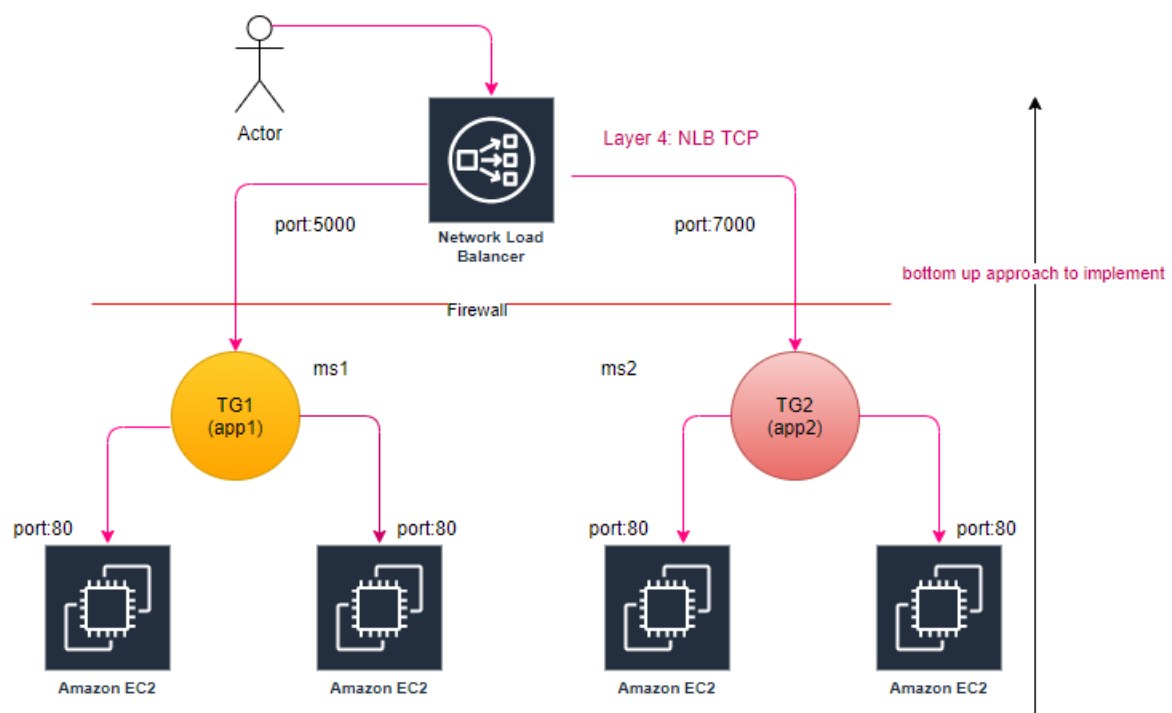
| Benefits                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Provide Secure Cloud Desktops for Remote Employees → AWS WorkSpaces</b><br>With Amazon WorkSpaces, your remote workers get a fast, responsive desktop of their choice that they can access anywhere, anytime, from any supported device                                                                                                           |
| <b>Enable Remote Learning and Working on legacy client server applications → AWS AppStream</b><br>fully managed application streaming service that allows students to access the applications they need on any computer, whether they're in the classroom, the library, a cafe, or at home                                                           |
| <b>Secure Content Collaboration → AWS WorkDocs</b><br>makes it easy to collaborate with others from anywhere and lets you easily share content, provide rich feedback, and collaboratively edit documents                                                                                                                                            |
| <b>Secure and Scalable Remote Network Access → AWS Client VPN</b><br>fully-managed, pay-as-you-go VPN service that elastically scales to support thousands of remote users. Client VPN is designed so your employees can access any company resource, both within AWS and on premises, from any location, with one console to manage all connections |

## Application Migration:

What is application migration?



ELB (Elastic Load Balancing): NLB uses burst algo to balance the traffic.



step1: Launch 2-2 instances per AZ

step2: Create TG for NLB (TCP), actual load balancing happens at TG level,  
**health check always be HTTP**

| Target groups (2) <a href="#">info</a>                      |                |                                 |      |          |             |               |  |
|-------------------------------------------------------------|----------------|---------------------------------|------|----------|-------------|---------------|--|
| <input type="text" value="Search or filter target groups"/> |                |                                 |      |          |             |               |  |
| <input type="checkbox"/>                                    | Name           | ARN                             | Port | Protocol | Target type | Load balancer |  |
| <input type="checkbox"/>                                    | TG-App-01-5000 | arn:aws:elasticloadbalancing... | 80   | TCP      | Instance    | NLB-01        |  |
| <input type="checkbox"/>                                    | TG-App-02-7000 | arn:aws:elasticloadbalancing... | 80   | TCP      | Instance    | NLB-01        |  |

step3: Register the Instaces (targets) with TG.

step4: Configure Routing→ Configure Target Group

Mapping-> corresponding to each subnet(routing) a seprate NIC is created.

**Listeners and routing** [info](#)

A listener is a process that checks for connection requests, using the protocol and port you configure. Traffic received by the listener is then routed per your specification.

▶ Listener TCP:5000
Remove

▼ Listener TCP:7000
Remove

Protocol
Port
Default action

TCP
7000
Forward to TG-App-02-7000 Target type: Instance TCP

1-65535
Create target group

These ports are immediately dropped as traffic routed

Validate:

- Is NLB end point is active.
- Is target group's targets are healthy
- This is also called reverse proxy?

You can host multiple applications behind single end point and redirect the traffic based on the port number.

### Tab: Integrated Services

- VPC Endpoint Service (AWSPrivateLink) Layer4
  - You can create your NLB as a service, for others to consume as a VPC Endpoint Service.
- Traffic Mirroring (Layer4)
- Global Configuration (pop)
- Config (record each activity in the account)

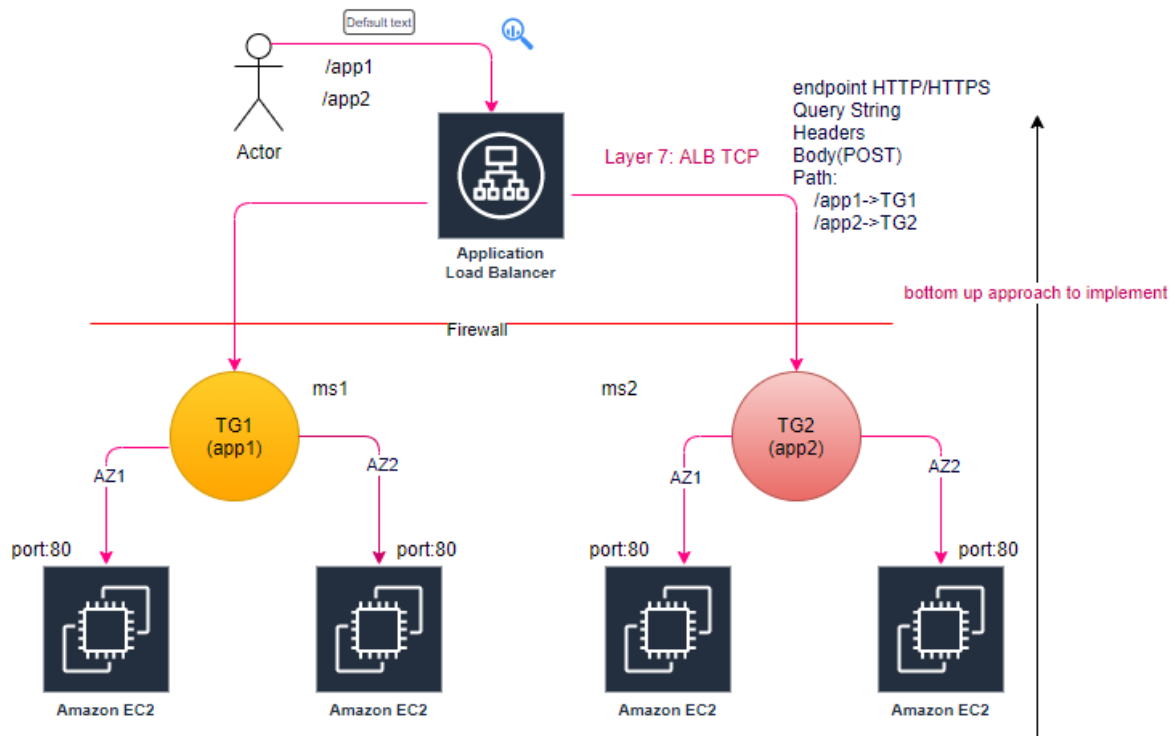
**Endpoint:** From consumer point of view

**Endpoint Service**-> publisher/provider

As a vpc end point can access within the VPC, while using endpoint service other can access.

All applications worked at region labels.

**ELB (Elastic Load Balancing):** ALB, uses round robin algo to balance the traffic.



step1: Launch 2-2 instances per AZ

step2: Create TG for NLB (TCP), actual load balancing happens at TG level,  
**health check always be HTTP**

Target groups (1/3) info

Search or filter target groups

|                                     | Name           | ARN                             | Port | Protocol | Target type | Load balancer |
|-------------------------------------|----------------|---------------------------------|------|----------|-------------|---------------|
| <input type="checkbox"/>            | TG-01          | arn:aws:elasticloadbalancing... | 80   | HTTP     | Instance    | -             |
| <input type="checkbox"/>            | TG-App-01-5000 | arn:aws:elasticloadbalancing... | 80   | TCP      | Instance    | NLB-01        |
| <input checked="" type="checkbox"/> | TG-App-02-7000 | arn:aws:elasticloadbalancing... | 80   | TCP      | Instance    | NLB-01        |

step3: Register the targets with TG.

step4: Configure Routing → Configure Target Group



Mapping-> corresponding to each subnet(routing) a separate NIC is created.

### Validate:

- Is ALB end point is active.
- Is target group's targets are healthy
- This is also called reverse proxy?

You can host multiple applications behind single end point and redirect the traffic based on the http/https, host-based routing (IP), path-based routing, http header-based, http-method based, Query String parameter-based, Source IP address CIDR based.

### Tab: Integrated Services

- AWS WAF (Layer7)
- Global Configuration (pop)
- Config (record each activity in the account)

## How to set Query based load balancing?

### Tab: Listener

step1: Click on view/edit rule

ALB-01 | HTTP:80 (2 rules)

▶ Rule limits for condition values, wildcards, and total rules.

Insert Rule

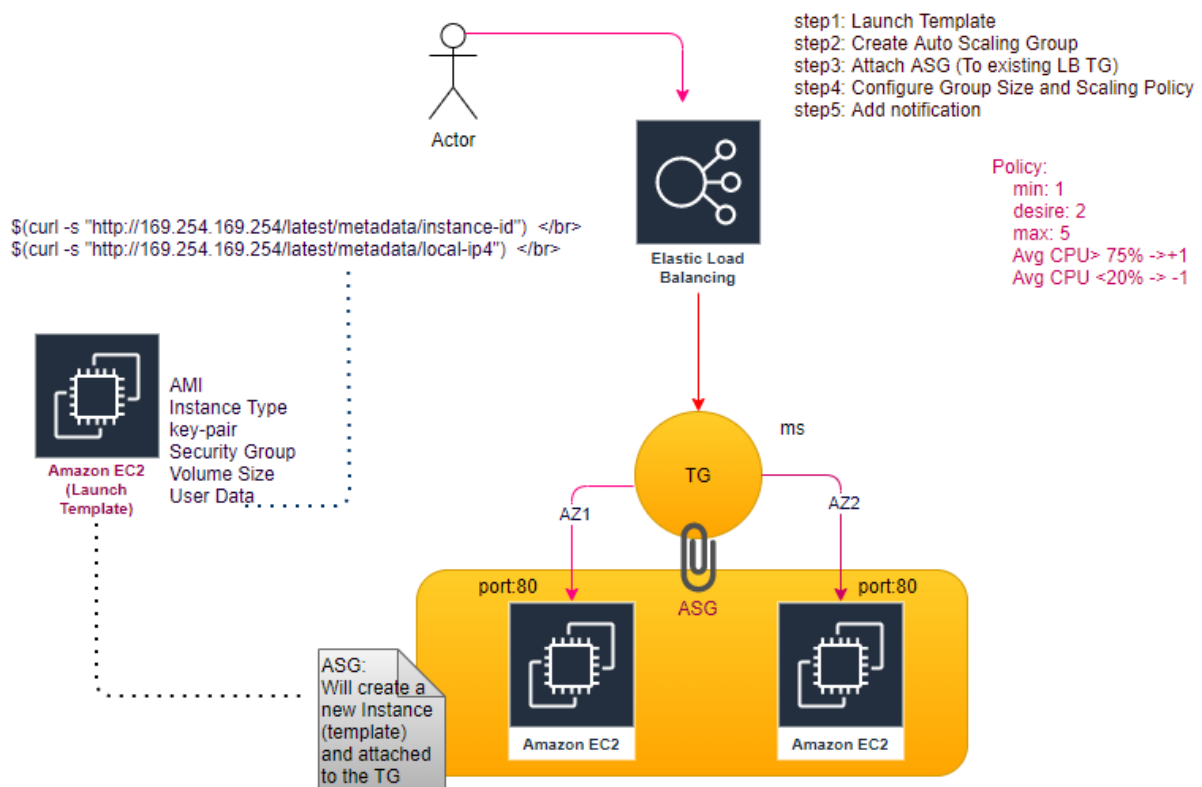
| RULE ID | IF (all match)                                                                                                                                      | THEN                                                                                                        |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 1       | <div>+ Add condition</div> <div>Host header...<br/>Path...<br/>Http header...<br/>Http request method...<br/>Query string...<br/>Source IP...</div> | <div>+ Add action</div> <div>THEN<br/>Forward to<br/>TG-01 : 1 (100%)<br/>Group-level stickiness: Off</div> |

last HTTP 80: default action  
This rule cannot be moved or deleted.

| # | Routing Option               | Route Client Request based on                            |
|---|------------------------------|----------------------------------------------------------|
| 1 | Host-based                   | HOST field in HTTP header                                |
| 2 | Path-based                   | URL path on HTTP header                                  |
| 3 | HTTP header-based            | Value of any standard field or custom HTTP header        |
| 4 | HTTP method-based            | Any standard or custom HTTP method                       |
| 5 | Query String parameter-based | Query String or Query Parameter                          |
| 6 | Source IP address CIDR-based | Source IP address CIDR from where the request originates |

All applications worked at region labels.

## Autoscaling (HPA):



step: Just map Auto Scaling group with the target group, along with policy, will automatically creates EC2 instance as per policy.

### Attach to an existing load balancer

Select the load balancers that you want to attach to your Auto Scaling group.

☒ Choose from your load balancer target groups  
This option allows you to attach Application, Network, or Gateway Load Balancers.

☐ Choose from Classic Load Balancers

Existing load balancer target groups  
Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups ▼

TG-01 | HTTP   
Application Load Balancer: ALB-01

## Scaling policies - *optional*

Choose whether to use a scaling policy to dynamically resize your Auto Scaling group to meet changes in demand. [Info](#)

- ☒ **Target tracking scaling policy**  
Choose a desired outcome and leave it to the scaling policy to add and remove capacity as needed to achieve that outcome.

☐ None

Scaling policy name

Target Tracking Policy - scale up

Metric type

|                                                    |   |
|----------------------------------------------------|---|
| Average CPU utilization                            | ▲ |
| Q Search metric types                              |   |
| Average CPU utilization                            |   |
| Average network in (bytes)                         |   |
| Average network out (bytes)                        |   |
| Application Load Balancer request count per target |   |

☐ Disable scale in to create only a scale-out policy

**AWS Beanstalk (PaaS):** Just handover your application to it (CloudFormation script executed internally).

- Auto Scaling
- Load Balancing
- Monitoring
- Logging
- Tracking
- Deployment Options
- Application Platforms
- Customization
- Web, Batch, Containers

**Platform**

Platform

-- Choose a platform --

.NET Core on Linux

.NET on Windows Server

Docker

GlassFish

Go

Java

Node.js

PHP

Python

Ruby

Using Beanstalk we can migrate our application, using the environments

**Select environment tier**

Amazon Elastic Beanstalk has two types of environment tiers to support different types of web applications. Web servers are standard applications that listen for and then process HTTP requests, typically over port 80. Workers are specialized applications that have a background processing task that listens for messages on an Amazon SQS queue. Worker applications post those messages to your application by using HTTP.

☒ **Web server environment**  
Run a website, web application, or web API that serves HTTP requests.  
[Learn more](#)

☐ **Worker environment**  
Run a worker application that processes long-running workloads on demand or performs tasks on a schedule.  
[Learn more](#)

It has its own limitation.

**CloudFormation (IaaS):** Stack to create AWS resources, Beanstalk creates a default (readymade) stack for given application, while using CloudFormation we can create customized (as per our requirement) stack.